Measuring Error-Control Code Performance in LEO Satellite Uplinks under Jamming

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Abstract

Low Earth Orbit (LEO) satellite networks play an increasingly important role in global communications but remain vulnerable to radio-frequency interference and jamming attacks. This work evaluates the performance of a widely used error-control coding scheme recommended by CCSDS standards, a Low-Density Parity-Check (LDPC) code with a polar-coded header, in LEO satellite uplink scenarios under jamming conditions. Both software simulations and software-defined radio (SDR) experiments were conducted to measure communication reliability using frame error rate (FER). Results show that both coding schemes experience performance degradation as jamming power increases. However, LDPC codes demonstrate more gradual degradation and maintain lower FER than Polar codes. SDR experiments show that LDPC-coded transmissions remain resilient to moderate jamming, with limited FER degradation observed for jamming-to-noise ratios below 0 dB. These findings suggest that LDPC codes provide strong robustness against interference jamming in the uplink and exhibit strong reasoning for their use in LEO satellite standards.

1 Introduction

LEO Satellites have gained popularity and interest in recent years for their ability to provide internet connectivity to users anywhere in the world, as long as they have a clear view of the sky. Compared with other satellite systems, LEO constellations offer lower latency and improved coverage, but they also introduce unique challenges due to the large number of satellites orbiting the Earth, which are readily accessible from the ground [1]. To maintain reliable communication, modern satellite and wireless systems rely heavily on error-correction codes, particularly channel-coding schemes such as low-density parity-check (LDPC) codes and polar codes. In recent geopolitical conflicts, the reliability of the connection enabled by these satellites has been tested as attackers attempt to jam and corrupt signals, preventing users from receiving internet services. LDPC codes are already used in satellite standards such as DVB-S2, while polar codes' quality of low complexity in both encoding and decoding have become important in 5G communications [2, 3].

It is important to understand the effect of jamming on the performance of these coding schemes, especially in the uplink, which presents specific reliability and security challenges, as the signal transmitted from a ground user to the satellite can be disrupted by radio-frequency interference. Satellite receivers may have limited ability to distinguish between noise, interference, and poor channel conditions, making jamming a particularly threatening security risk. Recent satellite security literature identifies these communication channels as vulnerable to jamming, spoofing, and interception, making resilience an important area of study [4]. This project studies the effects of jamming on LEO satellite uplink communications. The focus is on characterizing and comparing the performance of LDPC and Polar codes under interference. Using both software simulations and SDR-based over-the-air experiments, we analyze how jamming influences communication reliability and examine the relative robustness of LDPC-coded payloads and Polar-coded headers in satellite uplink scenarios.

2 Related Work

Error-control coding plays a central role in maintaining reliable communication in satellite networks. To address this, modern systems rely heavily on advanced channel coding techniques, with LDPC and polar codes among the most widely used. LDPC codes are already used in standards like 5G NR, WLAN, and DVB-S2, and prior research has explored their performance in satellite uplink (return link) scenarios. For example, studies using SC-FDMA-based uplink models show that NR LDPC tends to outperform other LDPC variants in terms of bit error rate, particularly for burst transmissions [5].

Polar codes have also gained attention for their theoretical ability to achieve channel capacity and relatively low complexity. They are used in 5G control channels and have also been studied for satellite communication. Much of the research on polar codes focuses on improving decoding performance, especially through belief propagation (BP) algorithms. Some recent work has even examined making these decoders more robust to hardware issues caused by the harsh space environment, showing that fault-tolerant decoding methods can improve reliability without degrading performance [6]. Comparative studies have also examined the relative performance of LDPC and Polar codes under conventional channel models such as AWGN and Rayleigh fading channels [3].

Satellite security researchers have recognized jamming to be a major threat to satellite and non-terrestrial networks. The vulnerability of satellite communication systems to jamming attacks highlights the growing importance of anti-jamming techniques for these networks [4]. However, most of this literature focused on characterizing jamming threats and mitigation schemes rather than evaluating the performance of specific error-control coding schemes under intentional interference, such as jamming. Moreover, most of the existing work examines performance under standard conditions, such as AWGN noise or fading channels [3, 5, 6]. There is much less work that considers intentional interference, especially in uplink scenarios. Also, while LDPC and polar codes have both been studied individually, there are relatively few direct comparisons between them under such adversarial conditions. In addition, relatively few studies combine software simulation with software-defined radio (SDR) experimentation to validate coding performance in realistic wireless environments [7]. Therefore, a significant gap remains in the study of the impact of jamming on modern error-control coding schemes in LEO uplink scenarios.

There remains a gap in understanding how modern LDPC and Polar codes perform under intentional jamming in LEO satellite uplinks. This project addresses this gap by evaluating both LDPC and Polar codes in a LEO satellite uplink scenario subjected to jamming. Through a combination of software simulations and SDR-based experiments, this work aims to provide a direct comparison of coding robustness under jamming and analyze the ability of these coding schemes in supporting reliable communication when the channel is actively disrupted.

3 Background

3.1 LDPC code

LDPC codes are a class of linear block codes defined by a sparse parity-check matrix H, which enables powerful error-correction performance with relatively low decoding complexity. In this work, we utilize the CCSDS short block length LDPC code with parameters $(128,64)$, corresponding to a code rate $R = 1/2$, where 64 information bits are encoded into a 128-bit codeword. The CCSDS $(128,64)$ LDPC code is designed for telecommand synchronization and channel coding in space communication systems, particularly for short-packet and low-latency uplink transmissions.

The LDPC encoder encodes the information block $u = [i_0, i_1, \ldots, i_{63}]$ into the codeword $x = [i_0, i_1, \ldots, i_{63}, p_0, p_1, \ldots, p_{63}]$ by generating parity-check bits such that the encoded codeword satisfies the parity-check constraint

$$Hx^T = 0 \tag{1}$$

where H is a sparse 64×128 parity-check matrix. The parity-check matrix can also be represented using a Tanner graph consisting of variable nodes and check nodes connected according to the nonzero entries of H. This graphical representation enables efficient iterative message-passing decoding algorithms.

Belief Propagation (BP) decoding is commonly used for LDPC codes, in which messages are iteratively exchanged between variable nodes and check nodes to estimate the transmitted bits. Due to the sparse structure of the parity-check matrix, BP decoding provides strong error-correction performance with relatively low decoding complexity.

3.2 Polar code

Polar codes are a type of linear block code that achieve channel polarization by separating bit channels into reliable and unreliable subchannels. Due to their low complexity, high reliability, and strong performance for short block lengths, polar codes are particularly suitable for 5G control channels and modern wireless communication systems.

The encoding process mainly consists of three steps: channel reliability estimation, bit assignment, and generator matrix encoding. Polar codes are constructed with a block length $N = 2^n$, where N must be a power of two due to the recursive structure of the channel polarization process. Although the encoder input vector has length N, only K positions carry information bits, resulting in a code rate of $R = K/N$.

During encoding, channel reliability estimation is first performed to separate the polarized bit channels into reliable and unreliable subchannels. The K information bits are assigned to the most reliable subchannels, while the remaining $N-K$ positions are assigned as frozen bits, which are typically fixed to zero. After constructing the input vector u, the encoding operation is performed using the generator matrix shown in Equation 2:

$$x = uF^{\otimes n} \tag{2}$$

where $u = [u_0, u_1, \ldots, u_{N-1}]$ is transformed into the encoded codeword $x = [x_0, x_1, \ldots, x_{N-1}]$. In Equation (1), $\otimes$ denotes the Kronecker product, where the kernel matrix F is defined as

$$F = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, \qquad F^{\otimes n} = F \otimes F^{\otimes (n-1)}. \tag{3}$$

Successive Cancellation (SC) decoding is the original decoding algorithm proposed for polar codes. SC decoding estimates each bit sequentially, using previously decoded bits and the received channel information, through a recursive, tree-like structure. Due to its relatively low computational complexity, SC decoding provides an efficient decoding method for polar codes.

3.3 Jamming attack

A jamming attack is a type of Denial-of-Service (DoS) attack that disrupts wireless networks by transmitting interfering signals in the same frequency band as legitimate communication. By flooding the channel with radio noise, attackers increase packet collisions, reduce throughput, and degrade network reliability and availability. Since wireless networks often operate in dynamic and mission-critical environments, jamming attacks can significantly affect communication performance and overall network functionality.

There are four common types of jamming attacks. The first is the constant-jamming attack, in which the attacker continuously transmits random signals to occupy the channel. The second is the deceptive jamming attack, which behaves similarly to spoofing by continuously sending seemingly legitimate packets to confuse receivers. The third is the random jamming attack, where the jammer alternates between active and idle states to conserve energy while still disrupting communication. The fourth is the reactive jamming attack, in which the attacker monitors the channel and only transmits interference when legitimate communication is detected.

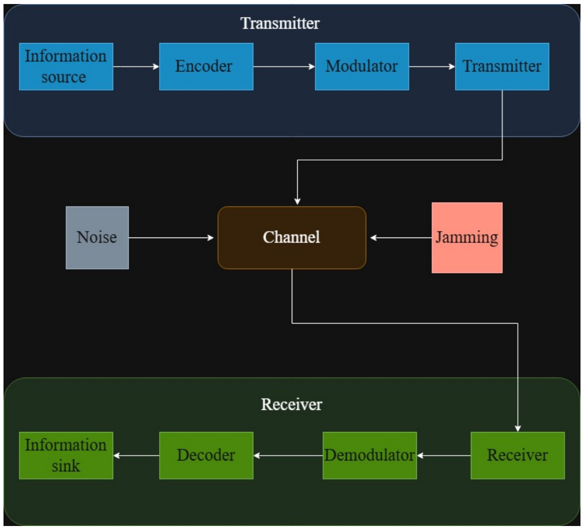

Figure 1: **The simulation framework evaluates coding performance under channel noise and jamming.** End-to-end communication system showing encoding, modulation, transmission, and decoding processes.

To defend against these attacks, several countermeasures have been proposed. Channel hopping and frequency hopping can mitigate the impact of jamming because the attacker must predict the hopping pattern in order to continuously interfere with communication. Multichannel receivers and error-correction codes (ECC) can increase the probability of successfully recovering corrupted messages in the presence of interference. In addition, encryption and authentication mechanisms can defend against deceptive jamming attacks by helping receivers distinguish legitimate packets from forged ones.

In this project, we focus on evaluating whether ECC techniques can effectively mitigate jamming attacks in practice. Specifically, we investigate whether LDPC and Polar codes can maintain reliable communication performance, and whether their practical performance matches the results predicted by theoretical analysis and simulations.

4 Methods

This project evaluates the performance of LDPC and Polar error-control codes in a jamming environment over a simulated low Earth orbit (LEO) satellite uplink system. The aim is to analyze and compare the effectiveness of these coding schemes under both simulated and over-the-air conditions by measuring frame error rate (FER) and bit error rate (BER).

4.1 Software Simulation Framework

A software simulation framework was developed to directly compare the performance of LDPC and Polar codes under Gaussian jamming. The software-based communication system is first implemented in Python. This pipeline is shown in Fig. 1 to establish a performance comparison between Low-Density Parity-Check (LDPC) codes and Polar codes. The simulated communication chain consists of source data generation, channel encoding, modulation, transmission through a noisy and jammed channel, demodulation, decoding, and error analysis.

The implemented LDPC and Polar codes are based on coding structures commonly used or proposed in modern LEO satellite communication systems [3]. The LDPC simulation uses a short random regular Gallager code with a 96-bit codeword length and an approximate rate of 1/2. Its sparse parity-check matrix has 48 parity-check equations, with each codeword bit participating in 3 checks and each check involving 6 bits. The message bits are first encoded using the LDPC coding scheme, and the resulting codewords are modulated using Binary Phase Shift Keying (BPSK). On the receiver side, the noisy received symbols are decoded using sum-product belief propagation (BP). The decoded message is then compared to the original transmitted message to compute FER and BER.

The Polar-code simulation follows a practical 5G-style design based on the standardized reliability sequence and successive cancellation decoding. The code uses a block length of $N = 1024$ bits and an information length of $K = 512$ bits, corresponding to a rate-1/2 code. Information-bit positions are selected according to the 5G reliability sequence, and encoding is performed using Arıkan's polar transform. The encoded bits are then modulated using BPSK and transmitted through the simulated channel. At the receiver, successive cancellation (SC) decoding is applied to recover the transmitted message. The decoded output is compared with the original information bits to compute FER and BER performance.

To evaluate performance under jamming, a Gaussian jammer is added to the AWGN channel. The jammer is modeled as an additional additive noise source whose power is varied to generate different jammer-to-noise ratios. Thus, the received signal contains both background channel noise and intentional Gaussian interference. Coding performance is compared as a function of the amount of interference.

4.2 SDR-Based Over-the-Air Experiments

An SDR-based 5G NTN-inspired uplink testbed was developed to evaluate coding performance under realistic transmission conditions. The over-the-air communication pipeline was built to emulate a LEO satellite uplink in the presence of jamming. The system architecture is based on 5G Non-Terrestrial Networks (NTN) concepts, with LDPC coding for uplink data transmission and Polar coding for control information [8]. The 5G NTN system uses 5G New Radio standards, and this pipeline adapts those concepts to a simplified SDR-based uplink emulation.

The transmission chain consists of a Zadoff-Chu (ZC) synchronization sequence, a Polar-coded header, guard symbols, and an LDPC-coded payload. ZC sequences are used because of their high performance in frame synchronization and timing alignment in noisy satellite channels. The receiver correlates the received signal with the known ZC sequence to find packet boundaries before decoding.

The packet header is protected using a short Polar code. The header contains control information required to interpret the payload, including the frame number, modulation identifier, code identifier, number of LDPC blocks, and payload length. A CRC-8 is appended to the header bits before Polar encoding. The resulting length-128 Polar codeword is mapped to QPSK symbols and transmitted before the LDPC payload. Although the design is inspired by Polar coding methods used for short control messages, it does not implement the full 5G NR Polar coding chain, such as NR-specific interleaving and rate matching [8]. At the receiver, the header is recovered using a maximum-likelihood search over a set of valid Polar-encoded header candidates rather than a standard SC or SCL Polar decoder.

A complete 5G NR Polar coding chain was not implemented because the objective of this work was to evaluate the effect of Gaussian jamming on a short coded packet structure. Therefore, a simplified Polar-coded header was suitable for SDR experimentation.

The payload is encoded using a CCSDS-inspired LDPC(128,64) code, corresponding to an approximate coding rate of 1/2 [9]. This coding structure was chosen since short LDPC codes are often proposed for satellite uplink reliability applications [9]. The payload encoder applies a sparse parity-check matrix, and payload decoding is performed using sum-product decoding.

For over-the-air transmission, the message bits are encoded and packetized using the frame structure described above. Both the Polar-coded header and LDPC-coded payload bits are modulated with Quadrature Phase Shift Keying (QPSK) before transmission. Root Raised Cosine (RRC) pulse shaping is applied before transmission through the SDR hardware. The modulated waveform is then upconverted to 915 MHz, corresponding to the resonant frequency of the SDR antennas used in the testbed. The transmitter SDR sends the signal to the receiver SDR, which samples the received signal at 1 MS/s.

At the receiver, the signal is downconverted to complex baseband. Coarse frequency correction is done by sending a tone before all transmissions and measuring the carrier frequency offset (CFO). Residual phase and frequency offset is estimated from the Zadoff-Chu preamble. An RRC matched filter is applied, and frame synchronization is performed using the preamble correlation. The preamble is also used to estimate and correct amplitude and phase distortion before QPSK demodulation, header recovery, and LDPC payload decoding. The header is recovered using a maximum-likelihood search over valid Polar-encoded header candidates, while the payload is decoded using the LDPC decoder.

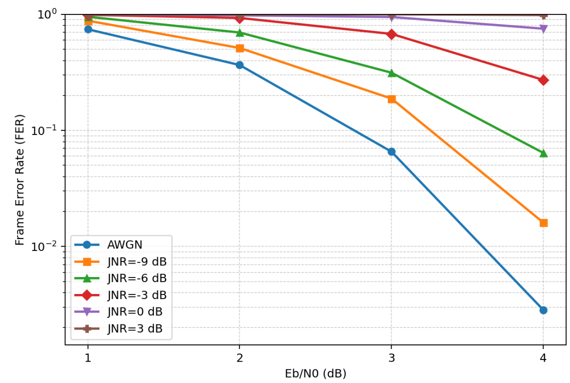

Figure 2: **LDPC performance degrades gradually under increasing jamming.** FER of the CCSDS (128,64) rate-1/2 LDPC code over an AWGN channel with varying JNRs. FER plots show that as jamming gain increases, FER performance slowly increases.

Jamming is introduced at the receiver by adding independent complex Gaussian noise to the received baseband samples. This emulates a Gaussian jammer while allowing accurate control of the jammer-to-signal ratio. The jammer power is set by a specified jammer-to-noise ratio relative to the estimated channel noise power. The decoder then uses the combined noise and jammer variances to compute log-likelihood ratios for accurate demodulation and decoding. The SDR approach is motivated by recent 5G NTN SDR prototyping platforms proposed for satellite communication validation [7].

5 Results

The LDPC simulations demonstrate robust performance under jamming, with FER increasing gradually as interference power rises. FER results for the rate 1/2 CCSDS (128,64) are shown in Fig. 2 for different bit power to channel noise ratios (E_b/N_0). Each FER point was simulated by running the LDPC code up to 200 errors. The best performance is achieved when jamming is absent, and the code is simulated over the AWGN only, as expected. As the jamming-to-channel noise ratio, measured in dB, increases, the FER slowly increases as expected. These initial results are key to our research group's goal of analyzing the performance of commonly used coding schemes in satellite uplinks under jamming. We will use these results to compare their performance to polar codes under jamming. This initial result also provides a baseline performance before we engage in the more complex task of simulating performance under jamming over the air using the SDRs.

Polar codes exhibit greater sensitivity to jamming than LDPC codes, with FER degrading more rapidly as interfer-

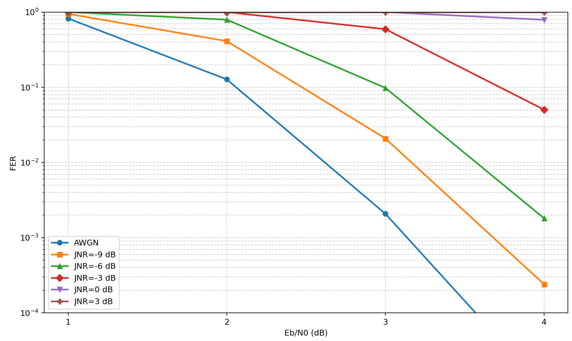

Figure 3: **Polar codes are more sensitive to jamming than LDPC codes.** FER of rate-1/2 Polar codes with SC decoding over an AWGN channel under Gaussian jamming. FER increases rapidly as JNR increases.

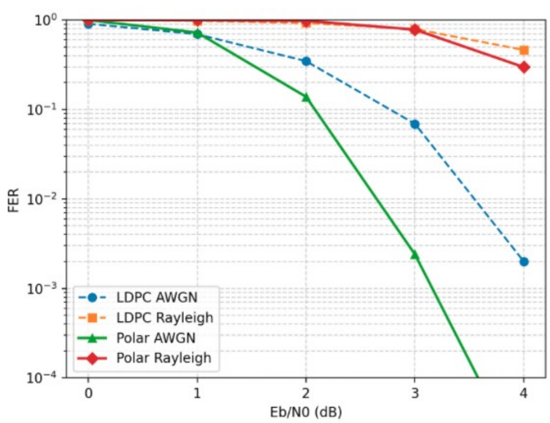

Figure 4: **Polar codes are more sensitive to channel variations than LDPC codes.** FER performance of Polar and LDPC codes over AWGN and Rayleigh fading channels. Polar codes exhibit a larger performance gap between the two channel models.

ence power increases. FER simulations for rate-1/2 Polar codes with SC decoding over an AWGN channel are shown in Fig. 3 under Gaussian jamming. Each FER point was obtained by simulating the code until 200 frame errors were observed. As expected, the lowest FER is achieved in the absence of jamming. Increasing the JNR results in a noticeable degradation in decoding performance, indicating reduced reliability under stronger interference. Compared with the LDPC results in Fig. 2, the Polar code experiences a steeper increase in FER, suggesting lower robustness to jamming.

To further investigate why Polar codes are more sensitive to jamming, we conducted additional simulations comparing Polar and LDPC codes over both AWGN and Rayleigh fading channels. Interestingly, Polar codes exhibit a significantly larger performance gap between the two channel models than LDPC codes. One possible explanation is that Polar codes rely on a predefined reliability ranking of subchannels for information-bit placement. In 5G systems, this reliability sequence is optimized for AWGN channels and remains fixed regardless of the actual channel conditions. Consequently, when the channel characteristics deviate from the AWGN assumption, the predefined reliability ordering becomes less accurate, leading to a greater performance loss. This suggests that Polar codes are generally more sensitive to changes in channel noise characteristics and, therefore, more susceptible to jamming-induced interference than LDPC codes.

Two SDRs were configured for over-the-air communication, with one SDR as the transmitter and the other as the receiver. Both SDRs are equipped with a band-pass filter before the transmit antenna centered at 915 MHz. Transmitted tones are at a frequency of 100kHz and are then upconverted to 915 MHz before they are transmitted. The communication protocol implemented for these simulations matches the structure described in the methods section. For the FER simulations, a random 64-bit information message is generated and encoded using the systematic LDPC generator matrix G, producing a 128-bit codeword. The coded bits are mapped to QPSK symbols, with two coded bits assigned to each complex symbol. A polar-coded header sequence mapped to QPSK symbols containing information about the payload (modulation/coding scheme) is prepended to the beginning of the payload. A Zadoff-Chu synchronization preamble is prepended to the full header and payload, and a zero-symbol tail is appended to the end of it. The resulting symbol sequence is pulse-shaped using a root-raised-cosine filter with roll-off factor $\beta = 0.5$, a span of 9 symbols, and 10 samples per symbol before transmission through the SDR.

Before the LDPC sweep, a single-tone calibration signal is transmitted to estimate the coarse carrier frequency offset between the transmitter and receiver. The receiver measures the tone frequency using an FFT and applies the resulting offset correction to subsequent received LDPC captures. Each received capture is DC-corrected, frequency-corrected, and passed through a matched root-raised-cosine filter. Frame synchronization is performed by correlating the received signal with the pulse-shaped Zadoff-Chu preamble. Around the strongest correlation peaks, the candidate with the highest correlation is selected.

The selected Zadoff-Chu preamble is then used to estimate the complex channel gain and residual phase slope across the frame. This phase slope provides a residual carrier frequency offset estimate, while the complex channel estimate is used for gain equalization. The phase correction and gain equalization are applied to the received QPSK payload symbols. The header is decoded using maximum-likelihood codebook search over valid polar-coded header candidates, rather than using a standard polar SC decoder. The equalized payload symbols are then converted into soft log-likelihood ratios (LLR) using the real and imaginary QPSK components and an estimated noise variance derived from the preamble er-

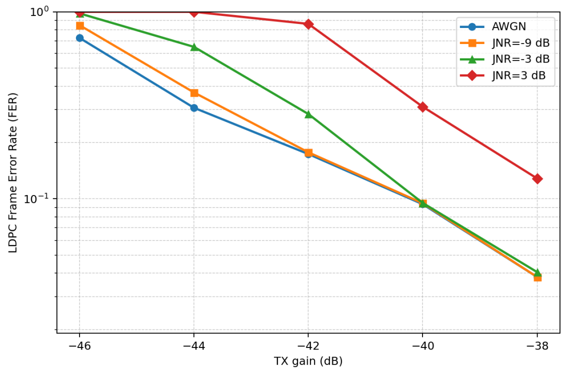

Figure 5: FER plot of over-the-air communications while sweeping over transmitter gain, keeping receiver gain fixed and running until 50 errors are achieved.

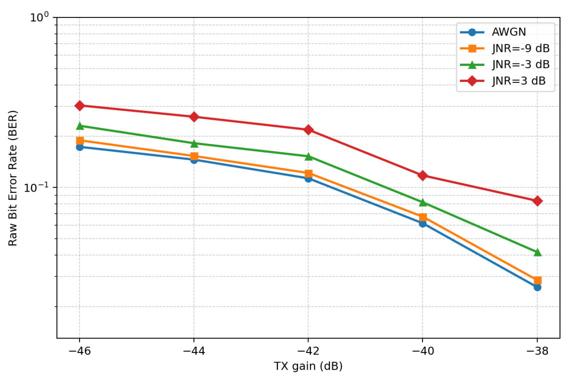

Figure 6: BER plot of over-the-air communications while sweeping over transmitter gain, keeping receiver gain fixed, and running until 50 errors are achieved.

ror vector magnitude. LDPC decoding is performed using a normalized min-sum message-passing decoder on the Tanner graph defined by the parity-check matrix H. The decoder iteratively exchanges messages between variable nodes and check nodes for up to 50 iterations. After each iteration, a hard decision is formed and checked against the parity constraint $H\hat{c}^T = 0$. A block is declared successfully decoded if the parity checks pass and the decoded codeword matches the transmitted codeword. The LDPC frame error rate is computed as the fraction of LDPC codewords that either fail to converge or decode incorrectly over the total codewords sent. In addition, the raw coded bit error rate is measured from hard QPSK decisions before LDPC decoding to separate RF/synchronization errors from LDPC decoding performance. The LDPC bit error rate is computed as the fraction of correctly decoded LDPC bits over the total bits sent.

The receiver gain was fixed at 40 dB for all SDR experiments, while the transmitter gain was swept from -46 dB to -38 dB. The frame error rate (FER), measured until 50 frame errors were observed for each JNR value, is shown in Fig. 5. At this receiver gain, increasing the transmit gain by only a few dB significantly improves FER, particularly in the presence of jamming. As expected, increasing the jammer-to-noise ratio generally increases FER. However, for JNR values below 0 dB, the jammer has little effect on FER compared with the no-jamming case, indicating that the LDPC code can tolerate low-power Gaussian jamming under these link conditions.

The raw bit error rate (BER) is shown in Fig. 6. Unlike FER, the BER shows visible degradation even for some JNR values below 0 dB, especially at higher transmit gains. This suggests that individual received bits are more sensitive to low-power jamming, while the LDPC decoder is still able to correct many of these errors before they result in full frame failures. Overall, the FER and BER results show that the LDPC-coded link is robust to Gaussian jamming when the jammer power is below the noise power, particularly when the transmit gain is sufficiently high.

The final observation from the results is that the measured header fail rate is zero across the tested TX gain and JNR values. Therefore, the polar-coded header decoder successfully recovered the header for all valid synchronized frames. This is likely due to the header being decoded using maximum-likelihood codebook search over valid polar-coded header candidates, rather than using a standard polar SC decoder.

6 Discussion/Limitations

The simulation results indicate that both LDPC and Polar codes experience performance degradation as jamming power increases. However, LDPC exhibits a more gradual FER increase, indicating greater robustness to interference. One possible reason is that LDPC's iterative belief-propagation decoder can better exploit redundancy across the codeword, while Polar codes with SC decoding are more susceptible to error propagation. This interpretation is supported by the AWGN–Rayleigh comparison, in which Polar codes exhibit a significantly larger performance gap than LDPC codes. Since the 5G Polar reliability sequence is optimized for AWGN channels, deviations from the assumed channel conditions can reduce the accuracy of the subchannel reliability ordering, making Polar codes more sensitive to channel variations and jamming.

LDPC decoding effectively mitigates the impact of low-power jamming, maintaining low FER even when BER begins to increase. The BER results from the SDR experiments provide additional insight into the system's response to interference. While BER increases for jammer-to-noise ratios below 0 dB at higher transmit gains, the corresponding FER remains close to the no-jamming case. This indicates that although the jammer introduces detectable bit errors, many are successfully corrected during LDPC decoding and therefore do not result in frame failures. The divergence between

BER and FER highlights the error-correction capability of the LDPC code under moderate interference conditions. Overall, the SDR results suggest that LDPC-coded communication remains highly reliable against jamming signals with JNR below 0 dB, particularly when the transmit gain is sufficiently large relative to the receiver gain.

The presented results should be interpreted with several limitations. First, the jamming model considered in this work is limited to additive Gaussian interference, whereas real-world adversaries may employ more sophisticated strategies such as reactive jamming or interference with different temporal and spectral characteristics. Second, in the SDR experiments, jamming was introduced by adding noise at the receiver side rather than transmitting interference from a dedicated jammer SDR. As a result, the experiments may not fully capture the physical-layer effects of over-the-air jamming. Finally, the SDR experiments were conducted in a controlled indoor environment and therefore do not fully represent the dynamics of real LEO satellite uplinks, where factors such as Doppler shifts, mobility, atmospheric effects, and varying propagation conditions may further influence communication performance.

Future work will focus on expanding the SDR testbed with a third dedicated SDR and an over-the-air jammer SDR to better emulate realistic satellite communication and interference scenarios. Additional studies will investigate a wider variety of jamming strategies, including reactive jamming, and evaluate Polar-coded payload performance in SDR simulations. These extensions will provide a more comprehensive understanding of error-control code performance and help identify more effective approaches for improving communication reliability in satellite uplinks.

7 Conclusion

This work evaluated the performance of LDPC and Polar codes in LEO satellite uplink communications under jamming using both software simulations and SDR experiments for LDPC codes. The software simulation results show that both coding schemes experience increasing frame error rates as interference increases, but LDPC codes exhibit more gradual degradation and greater robustness to jamming than Polar codes. Additional channel comparisons suggest that Polar codes are more sensitive to channel variations, which may contribute to their increased susceptibility to interference. The SDR experiments showed trends consistent with the software simulations and demonstrated that LDPC coding remains effective in practical over-the-air communications. While BER was affected by low-power jamming, FER remained relatively unchanged for jamming-to-noise ratios below 0 dB, indicating that the LDPC code was able to maintain reliable frame recovery under moderate interference. These results provide experimental evidence that LDPC coding can effectively improve communication robustness in the presence of jamming.

Contributions

Ava's main contributions to the research were finding the appropriate communication protocol for emulating LEO satellite uplinks, developing the LDPC code and min-sum decoder, and developing the code for the SDR over-the-air simulations. Ava's main contribution to the final paper was in the SDR experiment results section and their analysis in the discussion section.

Ting-Yu's main contributions to the research were developing the Polar code simulation framework and evaluating the performance of LDPC and Polar codes under different channel conditions and jamming scenarios. Ting-Yu's main contributions to the final paper were the background section, software simulation results section, and the corresponding analysis in the discussion section.

Thai's main contributions to the research include performing a literature review on modern LEO satellite communication systems and error-control coding techniques and assisting in improving the experimental methodology by incorporating concepts from current satellite communication standards into the system design. Thai's main contributions to the paper were the methods section describing the updated communication framework, the related works section, and the conclusion section summarizing the experimental results and key findings.

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